European Commission

Social Situation Monitor



Artificial Intelligence and Employment in the EU

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OVERVIEW

This research note examines the **employment impact of AI** in the **EU** between **2012 and 2022**, using both descriptive and regression analyses on 3-digit occupational data from the EU Labour Force Survey, along with an AI exposure indicator from Felten et al. (2021). The findings reveal that **AI exposure is highest** among highly educated professionals, particularly in analytical roles, which are mostly characterised by **non-routine cognitive** tasks. For instance, mathematicians, finance and legal professionals emerge as the most exposed. In contrast, occupations with the **lowest AI exposure** are generally manual and characterised by lower skill levels (e.g., mining and construction labourers). However, higher AI exposure does not necessarily equate to a higher risk of job substitution. Indeed, the **regression results** indicate that, on average, AI exposure is positively associated with employment growth, all else being equal.

KEY FINDINGS

#1

AI exposure is highest among highly educated professionals, particularly in analytical roles.

#2

AI exposure is positively associated with employment growth, suggesting a complementary relationship between AI and labour, at least for now, rather than substitution.

#3

The positive employment impact of AI is asymmetrically distributed across the EU; significant gains are seen only in countries characterised by stronger technological capabilities.

Highly skilled occupations appear to be more exposed to AI, with uncertain outcomes regarding augmentation or replacement, whereas manual occupations experience limited AI exposure

Ranking of occupations based on AI exposure

Ranking	ISCO 3-digit	ISCO 3-digit label	Type of prevalent task
		Top 10	
1	212	Mathematicians, Actuaries and Statisticians	NRCA
2	241	Finance Professionals	NRCA
3	261	Legal Professionals	NRCA
4	242	Administration Professionals	NRCA
5	431	Numerical Clerks	RC
6	231	University and Higher Education Teachers	NRCA
7	411	General Office Clerks	RC
8	122	Sales, Marketing and Development Managers	NRCP
9	251	Software and Applications Developers and Analysts	NRCA
10	233	Secondary Education Teachers	NRCP
		Bottom 10	
117	712	Building Finishers and Related Trades Workers	NRM
118	633	Subsistence Mixed Crop and Livestock Farmers	NRM
119	921	Agricultural, Forestry and Fishery Labourers	NRM
120	932	Manufacturing Labourers	RM
121	634	Subsistence Fishers, Hunters, Trappers and Gatherers	NRM
122	713	Painters, Building Structure Cleaners and Related Trades Workers	RM
123	911	Domestic, Hotel and Office Cleaners and Helpers	NRM
124	912	Vehicle, Window, Laundry and Other Hand Cleaning Workers	RM
125	631	Subsistence Crop Farmers	NRM
126	931	Mining and Construction Labourers	NRM

Source: Own elaboration based on Felten et al. (2021) and Lewandowski et al. (2020).

Notes: NRCA and NRCP stand for Non-Routine Cognitive Analytical and Non-Routine Cognitive Personal occupations, respectively. RM denotes Routine Manual, while NRM denotes Non-Routine Manual occupations.

To download the full research note, please visit the SSM website

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